

EBU6304 Software Engineering Group Project

2025-26

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First report

1. Requirements

1.1 Introduction

Teaching Assistants (TAs) play an important role in supporting teaching and academic activities. At BUPT International School, the current TA recruitment process is mainly handled through email and Excel files. As the number of applicants increases, this manual approach becomes difficult to manage and may lead to fragmented information, inconsistent records, and higher administrative workload.

To improve the efficiency and organisation of the recruitment process, this project aims to develop a Teaching Assistant Recruitment System that centralises applicant data and supports key recruitment activities. The system is intended to allow applicants to create and maintain profiles, upload CVs, browse available positions and submit applications, while enabling Module Organisers and administrators to review and manage applications more effectively.

The project follows Agile software development practices, with work organised into iterative cycles. In the first iteration phase, the team focuses on requirements gathering, stakeholder identification, backlog creation, and prototype development. These activities establish a foundation for the implementation and refinement of the system in later iterations.

This report summarises the work completed in the first iteration, including the requirement elicitation process, backlog prioritisation, and the planning of subsequent development iterations.

1.2 Stakeholders & Fact-Finding

Stakeholders

Teaching Assistant applicants are the main end users of the system. They need to create profiles, upload CVs, browse available jobs, apply for positions, and check their application status.

Module Organisers (MO) use the system to post jobs, review applications, and select suitable candidates. Their main concern is improving recruitment efficiency.

Administrators are responsible for managing the system and monitoring overall workload. They need accurate and well-organised information.

BUPT International School is the client organisation. It expects the system to reduce manual work and make the TA recruitment process more structured and efficient.

The **development team** is also an important stakeholder, as the system must be feasible to build, easy to maintain, and aligned with project constraints. Stakeholders may have different priorities and perspectives, so identifying them clearly is important in requirements engineering.

Fact-Finding

Interviews were carried out with Module Organisers and administrators to understand the current recruitment process and identify existing problems. This helped the team learn about issues such as lost CVs and disorganised email communication .

Questionnaires were used to gather feedback from TA applicants. This made it possible to collect opinions from multiple users efficiently. The survey showed that applicants wanted to track application status more easily and avoid entering the same personal information repeatedly .

Document analysis was also used by reviewing the current process based on email and Excel files. This helped the team understand how information is currently stored and managed, and why a more centralised system is needed .

Overall, using these techniques helped the team gather requirements from different perspectives and build a clearer understanding of user needs. This supports Agile development, where stakeholder feedback is important throughout the project .

1.3 Product Backlog

Each user story describes a feature from the perspective of a specific user role, including Teaching Assistant applicants, Module Organisers (MO), and Administrators.

The backlog was developed based on the system requirements and refined iteratively during the planning phase. Each user story follows the standard Agile format. This format ensures that each requirement is user-centered and clearly defines the value delivered by the system.

Every user story is defined with a structured set of attributes, including a unique Story ID (e.g., US-01) for traceability, a Story Name summarising the feature, a Description written in user story format, Acceptance Criteria that specify the conditions required for completion, Priority indicating the relative importance of the feature, Iteration Assignment mapping the story to a specific sprint, Estimation representing the expected implementation effort, and optional Notes providing additional constraints or clarifications. Together, these elements ensure that each requirement is clearly defined, traceable, and consistently interpreted throughout the development process.

Grouping user stories into modules helps to clearly separate concerns, making the system easier to understand, develop, and maintain. To improve clarity and maintainability, user stories are organised into modules based on responsibilities. The main modules identified in this project include:

- 1. User Management:** this module covers user authentication and account-related functionalities, such as registration, login, and logout (e.g., US-1 to US-6).
- 2. TA Profile Management:** this module includes creating, viewing, editing, and deleting applicant profiles, as well as managing CV uploads (e.g., US-07 to US-14).
- 3. Job Management:** this module focuses on browsing, searching, filtering, sorting, and viewing job listings (e.g., US-15 to US-18).
- 4. Application System:** this module allows applicants to apply for jobs and track their application status (e.g., US-19 to US-21).
- 5. MO Recruitment:** this module supports Module Organisers in posting jobs and selecting suitable applicants (e.g., US-24 to US-26).
- 6. Admin & AI Features:** this module includes administrative functionalities such as managing users, monitoring workload, and providing AI-assisted recommendations (e.g., US-22, US-23, US-27 to US-32).

Overall, the product backlog provides a structured and prioritised view of system requirements, supporting iterative development and effective project planning.

1.4 Iteration Planning

This project adopts an iterative development approach, dividing the development process into multiple stages, with each iteration lasting one week. In each iteration, the team implements a set of user stories and delivers a working version of the system, enabling continuous improvement and gradual system refinement.

According to the overall project schedule, the development is organised into four iterations:

Iteration 1 (22 Mar 2026 – 4 Apr 2026)

This iteration focuses on implementing the core system functionality, including user registration and login, applicant profile creation, CV upload, job browsing and application, as well as job posting and applicant selection by MO. This stage mainly covers high-priority

(Must) user stories, ensuring that the basic recruitment workflow can function correctly(e.g., US-01 to US-04, US-07, US-11, US-15, US-17, US-19, US-24 to US-26).

Iteration 2 (5 Apr 2026 – 12 Apr 2026)

Building on the core functionality, this iteration enhances the system by adding features such as profile editing and viewing, CV management, job search and filtering, application status tracking, and administrator functions. This stage mainly addresses medium-priority (Should) user stories, improving system completeness and user experience(e.g., US-05, US-06, US-08, US-09, US-12, US-13, US-16, US-18, US-20, US-28 to US-31).

Iteration 3 (13 Apr 2026 – 26 Apr 2026)

This iteration focuses on implementing extended features, including data export, skill matching, missing skill identification, and job recommendation. These features correspond to lower-priority (Could) user stories and aim to enhance the system's intelligence and usability(e.g., US-10, US-14, US-22, US-23, US-27, US-32).

Iteration 4 (27 Apr 2026 – 16 May 2026)

This final iteration concentrates on system integration and optimisation, including feature refinement, bug fixing, and final testing and demonstration preparation, ensuring system stability and readiness for delivery(e.g., US-21).

During the iteration planning process, the following principles were considered: User stories are scheduled based on their importance; Dependencies between features are taken into account when determining the implementation order; Core functionalities are prioritised to reduce development risks; Each iteration delivers a working version of the system.

Through this iteration planning, the system is developed progressively from core functionalities to advanced features, ensuring an organised development process and continuous improvement in system quality.

1.5 Prioritisation and Estimation

Prioritisation using MoSCoW Method

The MoSCoW prioritisation method is used to classify user stories in the Product Backlog based on their importance. This approach ensures that the development team focuses on delivering essential system functionalities first while less critical features can be implemented later if time permits.

The MoSCoW method consists of four categories:

Must have – essential features required for the system to function

Should have – important features but not critical

Could have – optional features that improve usability

Won't have – features not included in the current development cycle

From the Product Backlog, the majority of user stories are marked as **High priority**, indicating that the system development focuses primarily on core functionalities. These include key operations such as user management, application processing, and recruitment-related tasks, which are essential for the system to function.

Medium priority features provide additional support to improve usability and system efficiency, while low priority features mainly represent enhancements and optional improvements.

This prioritisation ensures that a functional system can be delivered early in the development process, with additional features added in later iterations.

Estimation Analysis

From the backlog:

Small tasks (1–3 story points) typically involve basic system operations such as simple user interactions, input handling, and straightforward functionalities. These tasks require minimal development effort and can be completed quickly.

Medium tasks (3–5 story points) involve moderate complexity, including data processing, interaction between components, and implementation of standard system features.

Large tasks (5–8 story points) represent complex functionalities that may involve multiple components, higher technical difficulty, or greater uncertainty. These tasks require more development effort and careful planning.

It is important to note that story points represent relative complexity rather than exact development time. For example, a task estimated at 8 story points is significantly more complex than a task estimated at 3 story points, rather than simply taking “more time”.

2. Supporting material

Interviews with Module Organisers and Admins revealed that almost all of participants struggled with lost CVs and messy email threads. This justified centralising applicant profiles and structured job postings as core "Must" requirements.

And the online survey of current TA applicants showed that most of them found it difficult to track their applications, leading to the inclusion of "Check application status" part. Another problem is repeatedly entering personal details. This strongly supported separating the "Applicant Profile" from the "Apply for Job" part, allowing one profile to be reused for multiple applications.

2.1 Interview Questions (MO & Admin)

To understand the current recruitment workflow and identify system requirements, interviews were conducted with Module Organisers (MO) and administrative staff.

Questions:

How do you currently manage TA recruitment?
What tools do you use (e.g., email, Excel)?
What problems do you face in handling applications?
How do you store and manage CVs?
How do you select suitable candidates?
What improvements would you like in a new system?

2.2 Interview Findings and Backlog Mapping

The interview results identified several key issues, which were directly translated into user stories in the product backlog.

Finding	Problem Description	Related User Stories
Disorganised email communication	Recruitment relies heavily on email threads, making information difficult to track	US-24, US-25 (MO job posting & applicant selection)
Lost or hard-to-find CVs	CVs are scattered across emails and files	US-07, US-11 (Profile & CV upload)
No centralised	No unified system to manage applicants	US-01–US-06 (User management),

applicant data		US-07–US-14 (Profile management)
Difficulty comparing candidates	Manual review is inefficient	US-26 (Applicant selection)
High administrative workload	Manual processes increase effort	Admin-related stories (US-22, US-23)

These findings justify the need for a structured and centralised recruitment system and support the prioritisation of core system features.

2.3 Questionnaire and Backlog Mapping

A questionnaire was distributed to Teaching Assistant applicants to collect broader user feedback.

Sample questions:

Do you find the current application process convenient?
Do you want to track your application status?
Do you need to enter the same information repeatedly?
What features would improve your experience?

Results and mapping:

User Feedback	Insight	Related User Stories
Difficult to track application status	Users lack transparency after applying	US-20, US-21 (Application tracking)
Repeated data entry is frustrating	Need reusable profile system	US-07–US-10 (Profile management)
Prefer centralised system	All functions should be integrated	Multiple core stories
Want clear process flow	Need better UI/UX and feedback	Prototype improvements

These results support key system functionalities and validate user-centered design decisions.

2.4 Observation & Document Analysis

A total of **12 responses** were collected from **Teaching Assistant applicants**.

The questionnaire aimed to understand user experience, identify key problems, and support requirement elicitation.

2.4.1 Key Results

1. Satisfaction with Current Recruitment Process

Very poor: 5 responses

Poor: 4 responses

Neutral: 2 responses

Good: 1 response

Very good: 0 responses

A bar chart would show that 75% of respondents rated the process as poor or very poor, indicating strong dissatisfaction.

This suggests that the current system fails to meet user expectations and requires significant improvement. Backlog impact: Supports development of core system features (all Must-have stories)

2. Main Problems Encountered (Multiple Choice)

Difficult to track application status: 10 responses

Repeated data entry: 9 responses

Unclear process: 7 responses

Slow response: 6 responses

Lost CVs: 5 responses

A horizontal bar chart would show "tracking status" and "repeated data entry" as the most significant issues.

These are the most critical pain points and directly influence system design priorities. Backlog mapping:

Application status tracking → US-20, US-21

Profile reuse → US-07–US-10

Structured workflow → Job & application modules

3. Importance of Application Status Tracking

Very important: 9

Important: 3

Neutral: 0

Not important: 0

A pie chart would show 100% of users consider this feature important, with the majority rating it as very important.

Application tracking is a critical requirement and must be prioritised. Backlog impact: Marked as Must-have feature (high priority)

4. Importance of Reusable Profile

Very important: 8

Important: 3

Neutral: 1

Not important: 0

A pie chart would show over 90% of users value profile reuse functionality.

Users strongly prefer avoiding repeated data entry, highlighting the need for a persistent profile system. Backlog impact: Profile management module (US-07–US-14)

5. Preferred System Type

Centralised system: 11

Email-based: 1

No preference: 0

A pie chart would show overwhelming preference for a centralised system.

This validates the decision to design a unified recruitment platform instead of relying on email. Backlog impact: Justifies entire system architecture

2.4.2 Overall Findings

Users are dissatisfied with the current process

Application tracking is a critical feature

Profile reuse significantly improves user experience

A centralised system is strongly preferred

These findings directly informed the product backlog and prioritisation decisions.

2.5 Prototype Feedback and Backlog Refinement

A low-fidelity prototype was presented to users to gather early feedback, consistent with Agile iterative development principles .

Feedback:

Users prefer a simple and intuitive interface

Navigation should be clear and minimal

Application status should be visible on dashboard

Important actions should be easily accessible

Impact on backlog:

Feedback	Design Improvement	Related User Stories
Simple UI	Reduce interface	UI-related refinement

preferred	complexity	
Need clear navigation	Improve page structure	All modules
Status visibility	Add dashboard/status feature	US-20, US-21
Easy access actions	Optimise workflow	Application & job modules

This feedback was used to refine both the prototype and the prioritisation of user stories in subsequent iterations.

Appendix: Questionnaire

Teaching Assistant Recruitment System Questionnaire

Section 1: Basic Information

1. What is your role?

Teaching Assistant applicant

Module Organiser

Administrator

Other: _____

2. Have you previously applied for a TA position?

Yes/No

Section 2: Current Experience

3. How do you currently apply for TA positions?

Email

Excel / Forms

Other: _____

4. How would you rate the current recruitment process?

(1 = Very poor, 5 = Very good)

5. What problems have you experienced? (Multiple choice)

Difficult to track application status

Repeatedly entering personal information

Unclear recruitment process

Slow response from organisers

Lost or missing CVs

Other: _____

Section 3: User Needs

6. Which features do you think are important? (Multiple choice)

Create and manage personal profile

Upload and manage CV

Browse available jobs

Apply for jobs online

Track application status

Receive notifications

Other: _____

7. How important is it to track your application status?

Very important

Important

Neutral

Not important

8. How important is it to reuse your personal profile (avoid re-entering data)?

Very important

Important

Neutral

Not important

Section 4: System Preferences

9. What type of system do you prefer?

Centralised online system

Email-based system

No preference

10. What interface style do you prefer?

Simple and minimal

Detailed with more options

No preference

Section 5: Open Feedback

11. What improvements would you like to see in a TA recruitment system?

12. Any additional comments or suggestions?
